
Amazon Macie

Amazon Macie for automated sensitive data discovery, classification, and protection in Amazon S3.

1. What is Amazon Macie?

Amazon Macie is a fully managed data security service that uses **machine learning** and **pattern matching** to:

- Discover and classify sensitive data (PII, PHI, credentials).
- Identify and alert on data security risks.
- Help with compliance (GDPR, HIPAA, PCI DSS).

 **Focus Area:** Amazon S3

2. Core Components

Component	Description
Sensitive Data Discovery Jobs	Scan selected S3 buckets to identify sensitive data types.
Findings	Alerts created when sensitive data is detected or configuration risks are found.
Custom Data Identifiers	User-defined regex patterns to find organization-specific data.

S3 Bucket Inventory	Visibility into all buckets: public access, encryption, etc.
Macie Dashboard	Visual overview of data risks and job results.
Integrations	EventBridge, Security Hub, CloudWatch for automation and response.

3. Hands-On Lab: Macie in Action

 **Goal:** Identify PII in a sample S3 bucket and generate findings.

Prerequisites

- AWS Account (Admin access)
 - One or more S3 buckets with sample data
 - IAM role with [AmazonMacieFullAccess](#)
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Steps

♦ Step 1: Enable Macie

- Go to AWS Console → Macie → Click **Enable Macie**
- Select the region

♦ Step 2: Review Bucket Inventory

- Macie will auto-detect S3 buckets

- Note any **public** or **unencrypted** buckets flagged

♦ **Step 3: Create a Sensitive Data Discovery Job**

1. Go to **Jobs** → **Create Job**
2. Select S3 buckets
3. Define scope (object tags, prefixes)
4. Choose managed data identifiers (e.g., credit cards, names, SSNs)
5. Schedule: One-time or recurring
6. Review and **Run Job**

♦ **Step 4: View Findings**

- Go to **Findings**
- Filter by severity, type (e.g., [SensitiveData:S3Object/Personal](#))
- Click on a finding → View object metadata

♦ **Step 5: (Optional) Create a Custom Identifier**

- Go to **Custom Data Identifiers**
- Define regex (e.g., Employee ID: [EMP-\[0-9\]{4}](#))
- Add keywords and context

- Use in your next job

♦ Step 6: Enable Integration

- Connect to **Amazon EventBridge**
 - Send findings to **AWS Security Hub** for centralized alerting
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4. Use Cases

Scenario	How Macie Helps
Financial Services Data Audit	Discover unencrypted credit card info in S3
Healthcare Compliance (HIPAA)	Identify and protect PHI in S3 buckets
Data Leakage Prevention	Alert when PII is exposed in public or misconfigured buckets
SaaS App with Customer Info	Scan logs/files uploaded by users for sensitive content

5. Key Learnings & Discussion

-  **Why only S3?** → Macie focuses on object-level data classification.
-  **ML + Pattern Matching** → Balances precision and speed.
-  **Custom Identifiers** → Needed for domain-specific formats.

-  **Continuous Monitoring** → Needed for dynamic environments.
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6. Classroom Questions

1. What types of data can Amazon Macie detect out of the box?
 2. How would you protect against sensitive data exposure beyond Macie?
 3. Compare Macie with Amazon GuardDuty.
 4. What are the limitations of pattern-based detection?
 5. How would you automate remediation based on Macie findings?
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7. Optional Extensions

- **Athena + Macie logs:** Query findings across multiple jobs.
 - **Macie + AWS Lambda:** Auto-quarantine sensitive data objects.
 - **Macie + WAF:** Block suspicious file uploads at the API level.
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8. Summary

 **Benefits**

 **Considerations**

Automated PII detection	Only scans Amazon S3
No infra to manage	Cost may increase with job frequency
Compliance-friendly	Limited to textual object content
Multi-account support	Some delay in job results



9. Sample Resources & Docs

- [Amazon Macie Official Docs](#)
 - [Pricing Guide](#)
 - [Workshop Labs](#)
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